

Soft Computing Question Bank

Unit 1

Chp 1 :

1. Explain Soft computing with its advantages and disadvantages
2. Explain hard computing , give difference between hard and soft computing
3. Differentiate between AI and Soft Computing?
4. Explain ANN with its process
5. Give application of Soft computing
6. What are the primary components of soft computing , Explain in detail
7. Give benefits and features if Soft computing ?
8. Explain the reasons to adapt Soft computing techniques with example

Chp 2 :

1. What is fuzzy logic ? give its components
2. Explain the importance of set theory in Fuzzy logic
3. Explain Fuzzy logic membership function and types
4. Give advantages and disadvantages of fuzzy logic
5. Explain fuzzy logic with its process
6. Differentiate between fuzzy logic and neural network
7. Explain Evolutionary Computation , give components of soft computing
8. What is genetic algorithm explain the steps in genetic algorithm
9. Give application of genetic algorithm in AI with its advantages and disadvantages
10. Explain the foundation of genetic algorithm , Give its operators

Unit 2

Chp 1 :

1. Explain Associative memory with its types
2. Explain adaptive Resonance theory with its key features
3. Explain Types of ART models with its application, Advantages and disadvantages
4. Explain Classification with example.
5. Explain Clustering with example
6. Explain Bayesian network with example
7. Explain Probabilistic reasoning with example
8. Explain application of probabilistic reasoning with its challenges

Chp 2:

1. Explain the types of Associative memories in brief
2. Explain Auto Associative memory network with its structure , properties and applications.
3. Explain Architecture of Auto associative memory with its training and testing algorithms.
4. Explain Hetro Associative memory network , with its structure , properties and applications.
5. Explain the architecture of hetro associative memory with its training and testing algorithm.
6. Explain Hopfield network with its architecture
7. Explain Bi directional associative memory, With its structure , properties and applications.

Unit 3 :

Chp1 :

1. Explain Simulated annealing . what are the advantages of using simulated annealing over gradient descent method in optimization ?
2. What is the key difference between Boltzmann machine and restricted Boltzmann machine.
3. Why are Boltzmann machines considered inefficient for large datasets ?
4. How does the Cauchy distribution differ from Gaussian distribution in terms of robustness to outliers
5. Explain Probabilistic Neural Network . gives advantages of using Probabilistic neural network for classification over a traditional neural network.
6. What distinguishes a cognitive network from a standard neural network?
7. How do cognitive networks simulate reasoning or decision-making?
8. What are the potential applications of cognition networks in fields like robotics or AI?
9. Explain Cellular Neural Network and Optical Neural Network.

Chp 2 :

1. Differentiate between fuzzy logic and Traditional Logic
2. Explain Classical Set theory
3. What is fuzzy logic give its characteristics of fuzzy logic
4. Explain the architecture of fuzzy logic system with membership function
5. Explain Set theory. Give types of classical set theory, with examples
6. Explain Operations on classical set theory
7. Explain properties of classical set theory
8. Define fuzzy set . Give operations on fuzzy set
9. Differentiate between classical set theory and fuzzy set theory. Give its applications
10. Give advantages and disadvantages of fuzzy logic

Unit 4

Chp 1:

1. How to find the cartesian product of 2 sets , explain with example.
2. What are the types of set.
3. Explain Classical relation and Fuzzy relation.
4. What is equivalence relation give its properties with example
5. Explain Fuzzification and defuzzification in detail (<https://www.tutorialspoint.com/difference-between-fuzzification-and-defuzzification>)

Chp 2:

1. What is genetic algorithm . Explain its basic terminologies
2. How genetic algorithm works
3. What is the purpose of Genetic algorithm
4. Differentiate between genetic algorithm and traditional algorithm .
5. What are the different selection operators ? Explain with example.
6. Explain cross over phase with its types , Give/list cross over operators
7. Explain the flow chart and architecture to run a genetic algorithm.
8. Give advantages and disadvantages of genetic algorithm

9. What are the applicable areas of genetic algorithm.
10. Explain Genetic algorithm and Traditional Optimum search method
11. Explain termination condition in genetic algorithm.